Data Dictionary

Voltkart · the tables you'll query

Overview

The data is generated from a fixed seed, so it is identical every time it is produced. The CSV files are in the `data/` folder, alongside a `load.sql` helper. The target engine is SQL Server.

dim_category

One row per category. This is a **self-referencing tree** of arbitrary depth — for example All Products → Electronics → Computers → Laptops → Gaming Laptops.

column	type	description
category_id	BIGINT	Unique category id
parent_category_id	BIGINT	The parent category. NULL for the single root
category_name	NVARCHAR	Category name

dim_employee

One row per employee. This is a **self-referencing org chart**: CEO → Regional Manager → Team Lead → Sales Rep. Orders are attributed to Sales Reps.

column	type	description
employee_id	BIGINT	Unique employee id
employee_name	NVARCHAR	Name
manager_id	BIGINT	Their manager. NULL for the CEO
role	NVARCHAR	CEO / Regional Manager / Team Lead / Sales Rep
region	NVARCHAR	North / South / East / West (All for CEO)

dim_customer

One row per customer.

column	type	description
customer_id	BIGINT	Unique customer id
customer_name	NVARCHAR	Name
city	NVARCHAR	City
segment	NVARCHAR	Consumer / SMB / Prime
signup_date	DATE	When the customer registered

dim_product

One row per product.

column	type	description
product_id	BIGINT	Unique product id
product_name	NVARCHAR	Product name
category_id	BIGINT	The product's leaf category
unit_price	DECIMAL	List price in INR
unit_cost	DECIMAL	Cost to Voltkart in INR
launch_date	DATE	When the product launched

fact_orders

One row per order (the header).

column	type	description
order_id	BIGINT	Unique order id
order_date	DATE	Date the order was placed
customer_id	BIGINT	The customer
sales_rep_id	BIGINT	The Sales Rep who owns the order
order_status	NVARCHAR	Completed / Cancelled / Returned

order_total	DECIMAL	Order value in INR (sum of its line amounts)

fact_order_items

One row per order line.

column	type	description
order_item_id	BIGINT	Unique line id
order_id	BIGINT	The order this line belongs to
product_id	BIGINT	The product
quantity	BIGINT	Units on the line
unit_price	DECIMAL	Price charged per unit (may differ slightly from list)
line_amount	DECIMAL	quantity multiplied by unit_price

stg_orders_incr

One row per order in the daily incremental batch (the 2025-03-16 feed). Same columns as `fact_orders`. It contains both updates to existing orders and brand-new orders, and it drives the incremental-load question.

cdc_product_changes

One row per product change captured from the source — a change-data-capture feed.

column	type	description
change_id	BIGINT	Unique change id
product_id	BIGINT	The product affected (unique across the feed)
operation	NVARCHAR	I = insert, U = update, D = delete
change_ts	NVARCHAR	When the change was captured
product_name, category_id, unit_price, unit_cost, launch_date	—	New values for I and U; last-known values for D

Things to keep in mind

> **Note** `dim_category.parent_category_id` is NULL at the root, and `dim_employee.manager_id` is NULL for the CEO. Anchor your recursive queries on those NULLs.

> **Tip** Use `order_status = 'Completed'` for revenue and performance questions unless a question says otherwise — Cancelled and Returned orders should not count as revenue.